



Artificial Intelligence for Software Developers

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Business Information Technology

Thesis

2024

Abstract

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Degree Bachelor of Business Administration
Title of the Report/Thesis Artificial Intelligence for Software Developers
Number of Pages and Appendices 29+3
The integration of Artificial Intelligence (AI) into software engineering is transforming the landscape of software development, enhancing the efficiency, productivity, and creativity of engineers. This thesis investigates the impact of AI, particularly the language model ChatGPT, on software engineering practices. It encompasses a detailed exploration of AI's evolution, current state, and future directions within the context of software development. The research begins with a historical overview of AI in software engineering, tracing its development from early rule-based systems to contemporary machine learning and natural language processing models. By reviewing a comprehensive array of scholarly articles, industry reports, and expert interviews, the study captures the multifaceted ways AI is reshaping software engineering. Key areas of impact include automated code generation, intelligent debugging, enhanced quality assurance, and dynamic documentation. A qualitative methodology is employed to analyze the contributions of AI, supplemented by practical experiments involving ChatGPT. These experiments include generating code snippets, providing documentation, and identifying bugs, demonstrating AI's practical applications and limitations. The findings highlight AI's ability to augment human capabilities, reduce repetitive tasks, and introduce innovative solutions to complex problems. A significant component of the thesis is a case study where ChatGPT is used to develop a simple portfolio website. This practical demonstration underscores AI's potential to assist in real-world software development tasks, from initial design to deployment. The results reveal that while AI can significantly aid in various stages of development, human oversight remains crucial to ensure the quality and relevance of the output. The thesis concludes with a discussion on the future implications of AI in software engineering, considering ethical, technical, and social dimensions. It emphasizes the need for ongoing research, interdisciplinary collaboration, and the development of guidelines to harness AI's benefits while mitigating risks. The study offers valuable insights into how AI can be effectively integrated into software engineering workflows, fostering a synergistic relationship between human expertise and machine intelligence. Overall, this thesis provides a comprehensive understanding of AI's transformative role in software engineering, presenting both theoretical insights and practical applications. It aims to serve as a foundational resource for researchers, practitioners, and policymakers interested in the future of AI-driven software development.
Keywords AI, Software, Developers, Future, DevelopWithAI, CodeWithAI, FutureOfAI

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1 Introduction

The integration of Artificial Intelligence (AI) into software engineering practices has emerged as a pivotal area of exploration, offering new avenues for streamlining development processes and improving outcomes. In recent years, AI technologies have increasingly permeated various facets of software development, from code generation to project management. This introduction sets the stage for the thesis's exploration of AI's role in revolutionizing software development practices. By delving into the intersection of human ingenuity and machine intelligence, this research aims to uncover the implications of AI for software engineers and project teams, paving the way for a deeper understanding of AI's transformative potential in the field.

1.1 Overview

Artificial Intelligence (AI) is rapidly reshaping the landscape of software engineering, offering unprecedented opportunities for innovation and efficiency. This thesis explores the integration of AI technologies into the software development process, aiming to elucidate their impact on the creation and maintenance of scalable applications. By investigating AI's role in enhancing the capabilities of software engineers and project teams, this research seeks to uncover the symbiotic relationship between human expertise and machine intelligence. Through an exploration of AI's historical evolution, current applications, and future prospects, this thesis endeavors to provide a comprehensive understanding of AI's transformative potential in software development. Additionally, for practical research purposes, this thesis includes QA testing processes with ChatGPT and in the final chapter, the development of a simple website for a barber shop is achieved through a series of four follow-up questions, demonstrating the practical implications of AI in real-world software development scenarios.

1.2 Research objective and question

In this research, our main objective is to find a solution about how Artificial Intelligence (AI) will help the software engineers to develop and maintain scalable applications and the future of Software Development. Nowadays, everybody is using some kind of AI, maybe they don't know they are using it, but they do. For example, Siri with the iPhone, so why can't software developers also use AI for their jobs? Now, the central part is first, we must understand what AI is and how can it help the developers. How can they use machine learning in their day-to-day job? What are the most popular AI algorithms and techniques? These are the main questions, and we will have answers later in this research process. The main goal is to understand the connection between Software Engineers and the usage of Artificial Intelligence.

1.3 Research Method

The research methodology to be employed will predominantly adopt a qualitative approach. This will involve an in-depth examination of scholarly articles, reports, and blogs, and the incorporation of personal experiences in Artificial Intelligence (AI), alongside insights from an experienced Software Engineer. Our study focuses on recent AI advancements, particularly delving into the application of ChatGPT across various software development scenarios, commencing from Chapter 3. Subsequently, in Chapter 4, an analysis of a third-party conducted survey sheds light on the efficacy of ChatGPT in enhancing software engineers' satisfaction and productivity levels. Towards the conclusion of the thesis, a demonstration follows wherein ChatGPT is utilized to generate code based on input, showcased through the development of a portfolio website as a practical assessment of its functionality.

2 Theoretical Background

The theoretical foundation of this thesis encompasses a comprehensive review of Artificial Intelligence (AI) and its historical evolution within the realm of software engineering. From its inception at Dartmouth College to contemporary advancements in deep learning and natural language processing, AI has undergone a remarkable journey of development. This section elucidates key historical milestones and theoretical frameworks that underpin AI's role in software development. Furthermore, the theoretical background delves into AI's future trajectory, exploring emerging trends and applications that hold promise for shaping the future of software engineering practices. Through a nuanced examination of AI's theoretical underpinnings, this research lays the groundwork for understanding its practical implications in software development.

2.1 Artificial Intelligence

In this section, we will delve into the fundamentals of Artificial Intelligence (AI), exploring its definition and scope. According to IBM (2021), Artificial Intelligence (AI), pertains to computer systems with the ability to execute intricate tasks that were traditionally exclusive to human capabilities, including reasoning, decision-making, and problem-solving.

Artificial Intelligence encompasses the capacity of machines to undertake tasks that typically demand human-like intelligence. (Lawretha 2023)

According to Stuart Russell and Peter Norvig (2019), it is a field of study that attempts to create intelligent machines by mimicking human intelligence.

What are the challenges in Artificial Intelligence?

According to Nilsson (1988) the main challenge in artificial intelligence is how to build an intelligent machine without being able to explain exactly why it does what it does and how it works.

This has led many researchers into exploring other approaches such as evolutionary algorithms, neural networks, genetic programming, and fuzzy logic. The problem with these methods, however, is they cannot be explained easily by humans or even computers themselves, so they have limited practical applications. (Hewlett Packard Enterprise 2019)

2.2 History of AI

In this section, we will explore the history of Artificial Intelligence (AI), tracing its origins and key milestones. We will examine the contributions of pioneers like John McCarthy and Alan Turing, and how advancements in computing technology have propelled the field forward.

Artificial Intelligence (AI) is a branch of computer science and engineering that focuses on the design, construction, operation, analysis, and application of intelligent agents or systems. According to McCarthy (2006), the term "Artificial Intelligence" was coined by John McCarthy in 1955 at Dartmouth College.

The groundwork for AI was laid earlier by Alan Turing, who wrote one of the first AI programs. This program, called 'The Imitation Game,' simulated a chess game between two players using logic to play against each other. Turing (1950) regarded this as a significant milestone in the history of artificial intelligence, demonstrating the potential for machines to perform tasks that typically require human intelligence.

This marked the beginning of the AI era. As computers became faster, cheaper, and more accessible, it became possible to improve machine learning algorithms. According to Bishop (2006), advancements in computing technology paved the way for enhancing these algorithms, leading to the sophisticated AI systems we have today.

2.3 Relevance of AI Tools in Software Development

Ensuring an optimal blend of human expertise and machine learning capabilities is essential for enhancing project management, testing, maintenance, revising risk-based requirements, and developing business-connected applications within the realm of software development. Presently, the predominant artificial intelligence systems utilized in software engineering predominantly rely on statistical or rule-based machine learning approaches. According to Batarseh, Mohod, Kumar and Bui (2018), the emergence of a machine learning method grounded in Artificial Neural Networks (ANN) or ChatGPT that claims the capacity to independently generate a new application purely through software coding would represent a significant milestone with the potential to reshape the global economy.

The integration of AI into the software engineering lifecycle has undergone rapid advancement in recent years. It is projected that by 2023, a substantial number of developers will adopt platforms or AI chatbots such as ChatGPT for pair programming. According to Puerto (2024) leveraging machine learning and AI technologies in software engineering processes can enhance efficiency, elevate quality standards, and reduce costs. The continuous enhancements in AI integration within the software development lifecycle have led to significant advantages across a diverse range of organizations.

3 Understanding ChatGPT

ChatGPT, a cutting-edge language model developed by OpenAI, represents a significant advancement in the field of artificial intelligence (AI). Leveraging deep learning techniques, ChatGPT can understand and generate human-like text, making it an influential tool in various applications that require natural language processing (NLP). This section explores the underlying architecture, development, functionality, and ethical considerations associated with ChatGPT, providing a comprehensive understanding of this sophisticated AI model. (Daniel Steele 2023)

3.1 History and Future of ChatGPT

“The development of ChatGPT can be traced back to the early days of artificial intelligence research” according to Daniel Steele (2023). In the nascent stages of computer systems' capabilities to comprehend and produce human language, a significant milestone emerged in the late 1990s. Researchers at Bell Labs pioneered the development of Cleverbot, an early endeavor in crafting a chatbot designed to emulate human dialogue. This system operated by employing an intricate framework of rules to formulate replies based on user interactions.

Later that a company called OpenAI, which is a nonprofit research company founded in 2015 by Elon Musk, Sam Altman, and Ilya Sutskever began to work on what would eventually become ChatGPT. (Ilya Sutskever 2015)

OpenAI's aim is to “progress digital intelligence in a manner most conducive to benefiting humanity, without being limited by a requirement for financial gain” wrote Ilya Sutskever to the Introduction page to the OpenAI website in 2015. It does this by funding researchers at institutions around the world who are developing general-purpose AI technologies for use in commercial applications. (Ilya Sutskever 2015)

According to Daniel (2023), the forthcoming advancements in language modeling promise an exciting trajectory in the years ahead. Researchers are actively investigating the capabilities of language models to innovate various forms of media, such as interactive narratives and gaming experiences. Moreover, efforts are underway to enhance models that can interpret and respond to human emotions, aiming to refine chatbots to be more interactive and human-like. The pivotal role of ChatGPT in language modeling continues to expand as researchers persistently explore novel methodologies and algorithms in natural language processing. Consequently, ChatGPT and other language models are poised to play a vital role in unlocking the immense potential embedded within human language.

4 Integration of ChatGPT in Software Development

This chapter explores the possible applications, advantages, and factors to consider when incorporating ChatGPT into software development processes. To gain insights into its utilization, we reference Alan S. Bluck's book (2023), illustrating ChatGPT's role in Java application development. Additionally, Cem Dilmegani's article (2024) outlines various real-world use cases of ChatGPT, shedding light on its integration within the software development domain.

4.1 Use Cases in Software Development

In this section, we will delve into the applications of ChatGPT and its relevance to Software Engineering. Various tasks are within the realm of a Software Engineer, as detailed in the subsequent chapter, where we will explore the utilization of ChatGPT in the capacity of a QA Tester and draw conclusions from that experience.

According to Cem (2024), "ChatGPT as a language model, can be used for a wide range of text-based use cases".

Coding Applications

According to Alan S. Bluck's (2023) publication, ChatGPT demonstrates the capacity to swiftly produce code in various programming languages. Nevertheless, precision and optimization may vary, necessitating precise prompts for optimal results. In a specific instance illustrated in Figure 1, I tasked ChatGPT with generating a file-reading algorithm for reading from "example.txt." The process was efficient and functional, albeit the task posed was not particularly challenging. However, in a subsequent section, we will enhance the development of a fundamental portfolio website for a barber shop utilized by ChatGPT to offer a more sophisticated demonstration, potentially presenting a higher level of complexity than the current task.

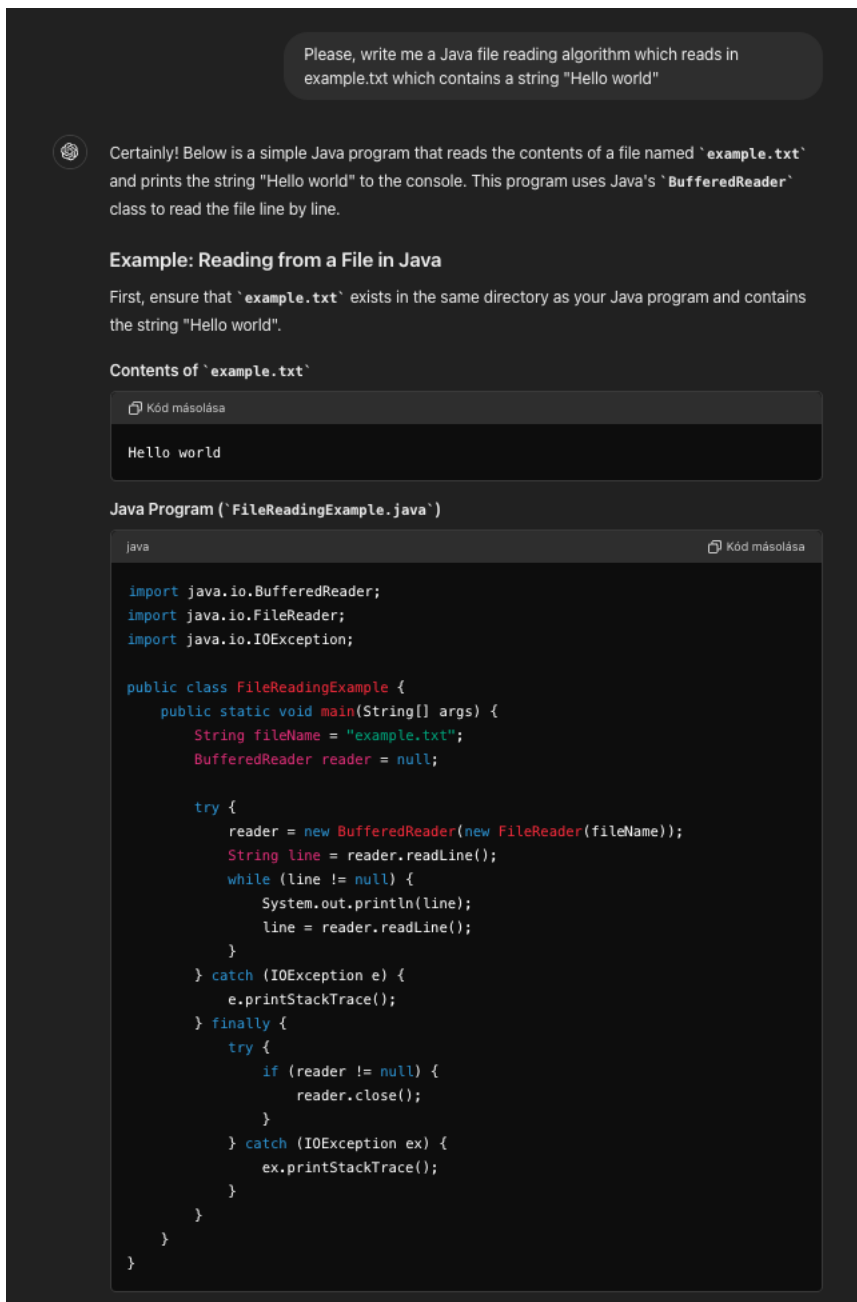


Figure 1. Java File Reading Example (Generated by ChatGPT)

Code Documentation

Within this segment, our focus remains on the Java code provided in Figure 1. While the code generated by ChatGPT in Figure 1 appears accurate, the absence of documentation poses a potential challenge. In a corporate setting, sharing code without accompanying documentation may hinder comprehension for other software developers. Comprehensive documentation holds significant value within large organizations, ensuring seamless collaboration among developers working on shared codebases.

In order to showcase this use case, I will request ChatGPT to revise the code and document the process, thereby assessing its performance in executing this task.

Can you add Code Documentation inside it?



Certainly! Below is the same Java program with added comments for better understanding and documentation:

Java Program with Documentation (`FileReadingExample.java`)

```

java
Kód másolása

import java.io.BufferedReader;
import java.io.FileReader;
import java.io.IOException;

/**
 * A simple Java program to read the contents of a file named "example.txt"
 * and print each line to the console.
 */
public class FileReadingExample {

    /**
     * The main method where the program execution begins.
     *
     * @param args Command-line arguments (not used in this program).
     */
    public static void main(String[] args) {
        // Name of the file to be read
        String fileName = "example.txt";

        // BufferedReader to read the file
        BufferedReader reader = null;

        try {
            // Initialize the BufferedReader with the specified file
            reader = new BufferedReader(new FileReader(fileName));

            // Read the first line from the file
            String line = reader.readLine();

            // Continue reading lines until the end of the file is reached
            while (line != null) {
                // Print the current line to the console
                System.out.println(line);

                // Read the next line
                line = reader.readLine();
            }
        } catch (IOException e) {
            // Print the stack trace if an IOException occurs
            e.printStackTrace();
        } finally {
            // Ensure the BufferedReader is closed to release system resources
            try {
                if (reader != null) {
                    reader.close();
                }
            } catch (IOException ex) {
                // Print the stack trace if an IOException occurs while closing the
                ex.printStackTrace();
            }
        }
    }
}

```

Figure 2. Code Documentation to the Java code (Generated by ChatGPT)

Conclusion

In this section, we explored the practical applications of ChatGPT within the realm of Software Engineering, focusing on its roles and effectiveness. The versatility of ChatGPT in generating code across various programming languages, as noted by Alan S. Bluck (2023), underscores its potential utility in coding applications. Our experiment with generating a file-reading algorithm illustrated the model's efficiency in producing functional code, though it highlighted the importance of precise prompts for achieving optimal results.

Furthermore, we examined the critical aspect of code documentation. The initial code generated by ChatGPT lacked documentation, posing a challenge for collaborative development environments. This emphasized the necessity of comprehensive documentation for ensuring seamless integration and understanding among developers in large organizations.

By revising the code and adding documentation through ChatGPT, we assessed its capability in enhancing code readability and maintainability. The documented code in Figure 2 serves as an example of how ChatGPT can be employed not only for code generation but also for improving documentation practices, thereby supporting better collaboration and code management.

In conclusion, ChatGPT demonstrates significant potential in assisting Software Engineers with coding and documentation tasks. While its ability to generate accurate and functional code is evident, the need for detailed prompts and comprehensive documentation remains crucial for maximizing its effectiveness. Future chapters will delve deeper into specific use cases, such as employing ChatGPT as a QA Tester, to further explore its applicability and draw more nuanced conclusions from these experiences.

4.2 Impact of ChatGPT in Software Development: Survey Analysis

In this segment, we shall delve into an analysis of a survey titled "Developers' Perceptions on the Impact of ChatGPT in Software Development: A Survey," (Thiago, Felipe, Paulo, Cuiyun, Jan, and Eduardo 2024). The survey targeted software developers, garnering responses from 207 individuals.

Question 1: How do developers perceive the effect of ChatGPT on their productivity while working on software development?

Based on the feedback received as you can see it on Figure 3, it is evident that ChatGPT enhances productivity in software development. While assessing productivity solely based on one metric proves challenging, approximately 50% of respondents believe that the utilization of ChatGPT has heightened their productivity levels in software development.

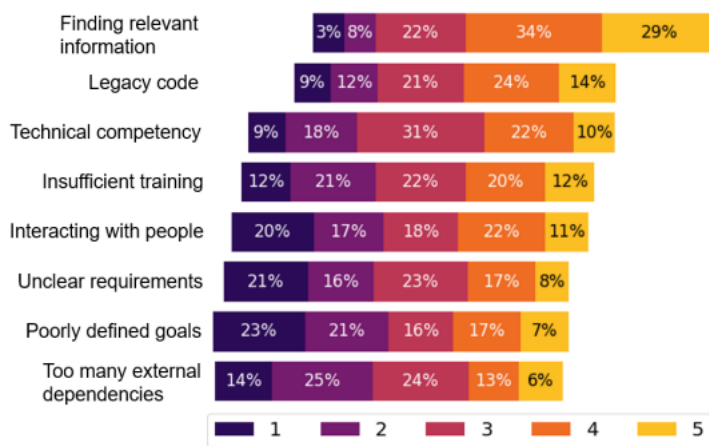


Figure 3. ChatGPT impact on developers productivity common challenges (Thiago et. al. 2024)

Question 2: How do developers feel that their satisfaction is impacted by ChatGPT?

The responses to this inquiry proved notably intriguing in my view. A majority of the audience acknowledged that ChatGPT had a positive effect on their satisfaction levels, although it did not seem to alter their workload intensity or stress levels. Nonetheless, there was a noticeable improvement in the efficiency of their routine tasks, with a remarkable 73% reporting a positive impact on goal achievement. It is noteworthy that many users have acquired valuable skills through ChatGPT in the workplace, resulting in a positive influence. Overall, user satisfaction with ChatGPT appears to be high as you can see it in Figure 4.

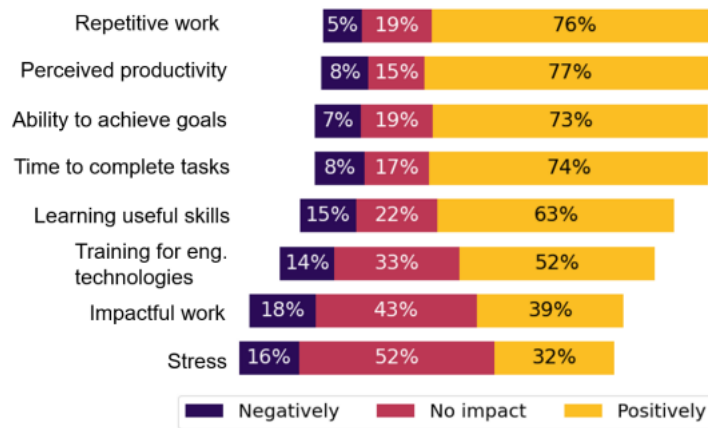


Figure 4. ChatGPT's influence on aspects of developers' job satisfaction (Thiago et. al. 2024)

Question 3: How does ChatGPT influence the time spent on specific software development tasks?

Following the assessment of satisfaction levels, productivity metrics, and an examination of the survey data, a decision was made to explore the impact of ChatGPT on task duration. Figure 4 illustrates a notable reduction in the time allocated to coding tasks attributed to the use of ChatGPT. As previously discussed, ChatGPT demonstrates proficiency in generating code efficiently with minimal user intervention required for specification and adjustments. While there is a decrease noted in the time allocated to code review and bug fixing, the impact is less pronounced. ChatGPT can assist in bug resolution, yet limitations in code complexity and length may restrict its applicability. Notably, debugging time for most users remains unchanged, a result that aligns with expectations.

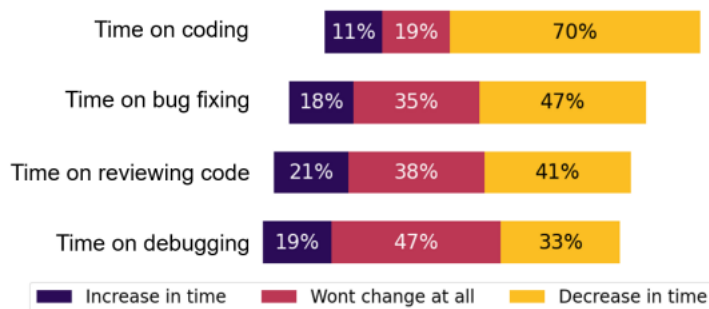


Figure 5. ChatGPT's influence on the Time Spent on Certain Task (Thiago et. al. 2024)

Conclusion

In conclusion, the survey analysis focusing on the impact of ChatGPT in software development has unveiled several significant findings. Primarily, a noteworthy proportion of developers perceive

ChatGPT as a catalyst for enhancing productivity in software development, as approximately half of the respondents have reported a positive impact. This underscores the tool's efficacy in simplifying processes and facilitating task completion.

Furthermore, the survey outcomes underscore a positive relationship between ChatGPT usage and developers' satisfaction levels. While ChatGPT may not directly alleviate workload intensity or stress, it does contribute to heightened efficiency in routine tasks, thereby aiding in goal attainment. Notably, users have also highlighted the acquisition of valuable skills through their interactions with ChatGPT, further enriching their job satisfaction.

Moreover, the analysis demonstrates that ChatGPT influences the time allocated to specific software development tasks. Particularly, there is a decrease in coding time, attributed to ChatGPT's ability to generate code swiftly. However, its impact on code review and bug fixing appears less pronounced, indicating certain limitations, especially concerning code complexity and length.

Overall, the results suggest that ChatGPT plays a constructive role in software development by bolstering productivity, user satisfaction, and efficiency in select tasks. Recognizing both the strengths and constraints of ChatGPT is crucial to optimizing its effectiveness within the development process. As ChatGPT technology advances, further research and exploration will be essential to fully grasp its potential and enhance its seamless integration into software development workflows.

5 Best Practices for using ChatGPT in Software Development

This chapter delineates the optimal practices for leveraging ChatGPT. While the language model presents numerous promising opportunities, it is imperative to prioritize accuracy and relevance in the generated responses. Adhering to these best practices guarantees that the model is trained on superior data, yields precise outcomes, and is seamlessly incorporated into the development workflow.

According to Jelena (28 April 2023), the following best practices are deemed least significant in the utilization of ChatGPT for software development objectives.

- **Understand the limitations of ChatGPT:** ChatGPT, while proficient in generating natural language responses, is not flawless and may occasionally yield inaccurate or irrelevant outputs. It is crucial to comprehend these limitations and meticulously assess the responses to ensure accuracy and relevance.
- **Choose the right training data:** The efficacy of the ChatGPT model heavily relies on the quality of the training data. Hence, selecting pertinent training data that aligns with the specific use case and encompasses a diverse array of examples is imperative.
- **Regularly retrain the model:** ChatGPT models can become outdated as language evolves and new data emerges. Consistent retraining with updated data is essential to uphold the model's ability to provide factual and pertinent responses continually.
- **Use context to improve accuracy:** Context plays a pivotal role in enhancing natural language comprehension. Providing comprehensive context to ChatGPT can significantly boost its accuracy. Hence, it is advisable to furnish the model with as much context as possible when inputting data.
- **Test and refine the model:** Thorough testing and refinement are indispensable for ensuring that ChatGPT furnishes valid and relevant responses. Testing the model with diverse inputs and refining it based on feedback are essential steps to enhance its performance progressively.
- **Ensure ethical and responsible use of ChatGPT:** Upholding ethical and responsible practices in ChatGPT usage necessitates developers and stakeholders to meticulously assess the potential implications of their applications. Establishing clear guidelines for ChatGPT usage, testing and monitoring the model for accuracy and bias, and incorporating diverse perspectives and feedback during the development process are crucial measures to mitigate adverse consequences.

Drawing upon established best practices, software developers adeptly harness language models and ChatGPT to enhance productivity levels and craft applications that are intuitive and seamlessly natural in their interactions.

5.1 Quality Assurance and Testing

This chapter explores how ChatGPT can enhance software testers' work, focusing on key areas such as test case scenario generation, bug reproduction and reporting, test data generation, exploratory testing, test automation assistance, user experience testing, and continuous learning. Based on a Yasin Bulgan's (Yasin 29 November 2023) article, he conducted plenty of testing scenarios with ChatGPT and explored the potential benefits for QA professionals.

Test Data Generation

I aim to assess and comprehend the speed and accuracy with which ChatGPT can generate test data for a MySQL database.

My questions towards ChatGPT were the following: "Can you generate random data for MySQL database columns: username, first name, last name, registration date, city, and level?"

Certainly! Here's a script to generate random data for a MySQL database with the columns: `username`, `first\_name`, `last\_name`, `registration\_date`, `city`, and `level`. The script will create 100 random entries for demonstration purposes.

```
sql Kód másolása

CREATE TABLE users (
  id INT AUTO_INCREMENT PRIMARY KEY,
  username VARCHAR(50) NOT NULL,
  first_name VARCHAR(50) NOT NULL,
  last_name VARCHAR(50) NOT NULL,
  registration_date DATE NOT NULL,
  city VARCHAR(50) NOT NULL,
  level INT NOT NULL
);

DELIMITER //

CREATE PROCEDURE generate_random_users()
BEGIN
  DECLARE i INT DEFAULT 0;
  DECLARE username VARCHAR(50);
  DECLARE first_name VARCHAR(50);
  DECLARE last_name VARCHAR(50);
  DECLARE registration_date DATE;
  DECLARE city VARCHAR(50);
  DECLARE level INT;

  WHILE i < 100 DO
    SET username = CONCAT('user', FLOOR(RAND() * 10000));
    SET first_name = ELT(FLOOR(RAND() * 10) + 1, 'John', 'Jane', 'Alice', 'Bob');
    SET last_name = ELT(FLOOR(RAND() * 10) + 1, 'Smith', 'Johnson', 'Williams');
    SET registration_date = DATE_ADD('2020-01-01', INTERVAL FLOOR(RAND() * 1600) DAY);
    SET city = ELT(FLOOR(RAND() * 10) + 1, 'New York', 'Los Angeles', 'Chicago');
    SET level = FLOOR(RAND() * 5) + 1;

    INSERT INTO users (username, first_name, last_name, registration_date, city, level)
    VALUES (username, first_name, last_name, registration_date, city, level);

    SET i = i + 1;
  END WHILE;
END //

DELIMITER ;

CALL generate_random_users();
```

Figure 6. Test Data for MySQL Database (Generated by ChatGPT)

The outcome was unexpectedly remarkable as ChatGPT successfully produced a functional MySQL code that creates 100 random entries for my database. Furthermore, it comprehensively explained the script's functionality and guided its execution.

Debugging

The upcoming scenario involves fundamental debugging, with the primary inquiry focusing on ChatGPT's ability to rectify faulty JavaScript code. Within Figure 2, an erroneous JavaScript code is presented, featuring typographical errors such as missing characters and a loop exceeding its bounds.

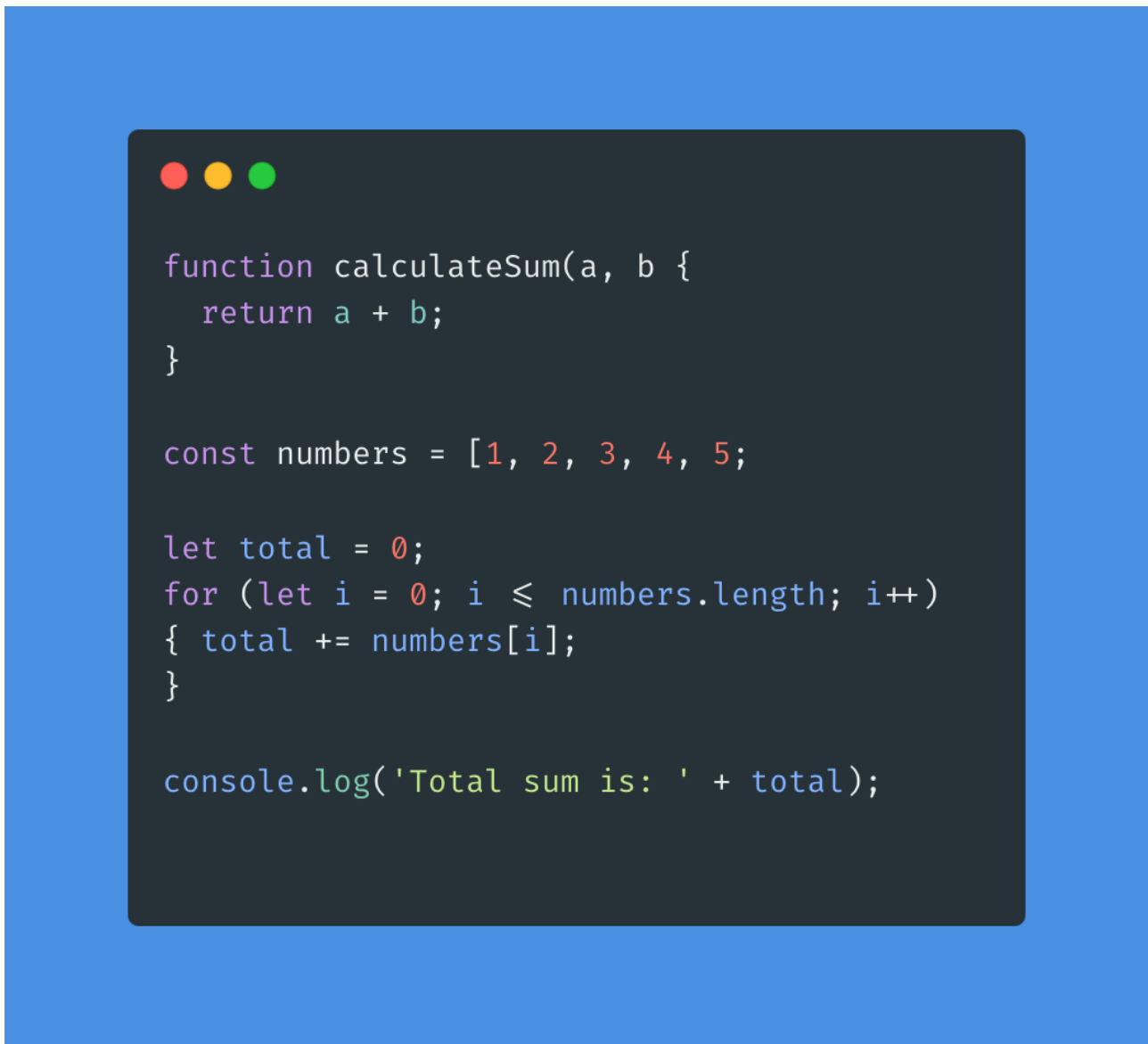


Figure 7. Wrong JavaScript code example

The malfunctioning code prompted my interest in ChatGPT's debugging capabilities. Upon providing the flawed code for analysis, the outcome was remarkably impressive.

Certainly! Here is the corrected version of the code:

javascript

 Kód másolása

```
function calculateSum(a, b) {  
  return a + b;  
}  
  
const numbers = [1, 2, 3, 4, 5];  
  
let total = 0;  
for (let i = 0; i < numbers.length; i++) { // Fix the loop condition  
  total += numbers[i];  
}  
  
console.log('Total sum is: ' + total);
```

Summary of Fixes:

1. Fix the function declaration by adding the missing parenthesis:

javascript

 Kód másolása

```
function calculateSum(a, b) {
```

2. Close the array declaration properly with a closing bracket:

javascript

 Kód másolása

```
const numbers = [1, 2, 3, 4, 5];
```

3. Adjust the loop condition to avoid going out of bounds:

javascript

 Kód másolása

```
for (let i = 0; i < numbers.length; i++) {
```

Now the code is corrected and will output the total sum of the numbers in the array correctly.

Figure 8. Wrong JavaScript Code Fix (Generated by ChatGPT)

As depicted in Figure 3, ChatGPT swiftly rectified the JavaScript code and furnished a concise summary alongside an explanation of the modifications. Such expeditious error identification and elucidation greatly streamline the debugging process for Software Developers.

Conclusion

The findings from these scenarios highlight the potential of ChatGPT as a beneficial tool for Quality Assurance professionals and software developers. Through streamlining test data generation and debugging processes, ChatGPT can enhance productivity and precision in software testing and development. The conclusions drawn from these assessments confirm that the integration of AI tools like ChatGPT into the QA workflow can lead to substantial improvements in quality assurance methodologies.

6 Building a Website with ChatGPT

In this section, we will employ the ChatGPT tool, which is available at no cost to assist us in creating a basic portfolio website. This endeavor aims to assess the feasibility of utilizing AI for developing a simple website. It is important to acknowledge that while AI can aid in the process, achieving a flawless and fully functional webpage may still require human intervention.

6.1 Folder structure generation

In this section, we are seeking ChatGPT's assistance in devising an exemplary folder structure for the project and furnishing detailed explanations to initiate the website development process. As depicted in Figure 9, ChatGPT is offering guidance on establishing our directories to commence website development.

Sure! Here's a basic folder structure for a simple portfolio website for a hairdresser company:

```

markdown
Copy code

hairdresser-portfolio
|
├── index.html
├── css
│   └── styles.css
├── images
│   ├── logo.png
│   └── portfolio
│       ├── image1.jpg
│       ├── image2.jpg
│       └── ...
└── js
    └── script.js
  
```

Explanation of each component:

- `index.html`: This is the main HTML file that will serve as the homepage of your website.
- `css`: This folder contains all the CSS files for styling your website.
 - `styles.css`: This file will contain the CSS styles for your website.
- `images`: This folder stores all the images used on your website.
 - `logo.png`: The logo image of the hairdresser company.
 - `portfolio`: This subfolder contains images of the hairdresser's portfolio, such as hairstyle examples, salon interior, etc.
- `js`: This folder contains JavaScript files.
 - `script.js`: This file will contain any JavaScript code needed for interactive elements on your website, such as sliders, form validation, etc.

You can expand upon this structure as needed, adding additional HTML files for different pages, more CSS files for styling variations, and more JavaScript files for additional functionality.

Figure 9. Folder structure

6.2 Website creation inputs

Within this chapter, I have compiled the four inputs and outputs detailing how ChatGPT facilitated and provided results throughout the website creation process. Witnessing the language model's capability to assist individuals, irrespective of their programming expertise, has been a rewarding experience, particularly due to the clear and comprehensive explanations provided. Please refer to Table 1 for a comprehensive overview of the inputs pertinent to our website development endeavour.

Table 1. Inputs for the website creation (ChatGPT)

	Input	Output
<i>First step:</i>	I would love to create a simple portfolio website for a hair-dresser company. Please create me the basic folder structure.	Sure! Here's a basic folder structure for a simple portfolio website for a hairdresser company See it in Figure 1
<i>Second step:</i>	Please, let me help to create the content of the index.html. It should be a basic hair-dresser-portfolio website, which should contains a name which is MR.Barber and pricing of course. Also, include a footer which contains the copyright + example contact	Certainly! Below is a basic example of index.html for your hairdresser portfolio website See it in Figure 3
<i>Third step:</i>	Please, use a minimalistic design for the styling, and be careful with the user-experience. Try to use mobile-first methodology.	Of course! Here's a revised version of the CSS with a minimalistic design and mobile-first approach See it in Figure 4
<i>Final step:</i>	Can you create for me some functionality inside the script.js? To have wonderful scroll bar with smooth animations.	Certainly! Below is an example of how you can implement a smooth scrolling functionality with animations using JavaScript See it in Figure 5

6.3 Website content

The language model efficiently generated the content for the website, providing the entire text of the index.html file that can be seen in Figure 10, which I could seamlessly copy and paste into the code editor. It not only swiftly responded to my inputs but also incorporated additional details and sections as per my request. This ensured that the website's content was comprehensive and tailored to my specifications. Overall, the process was smooth, and the language model's responsiveness significantly expedited the website development process.

```

styles.css  JS  script.js  index.html X
index.html > html
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>MR. Barber - Hairdresser Portfolio</title>
7      <link rel="stylesheet" href="css/styles.css">
8  </head>
9  <body>
10     <header>
11         <h1>Welcome to MR. Barber</h1>
12         <nav>
13             <ul>
14                 <li><a href="#services">Services</a></li>
15                 <li><a href="#portfolio">Portfolio</a></li>
16                 <li><a href="#contact">Contact</a></li>
17             </ul>
18         </nav>
19     </header>
20
21     <section id="services">
22         <h2>Our Services</h2>
23         <ul>
24             <li>Haircut - $25</li>
25             <li>Shave - $20</li>
26             <li>Beard Trim - $15</li>
27             <!-- Add more services here -->
28         </ul>
29     </section>
30
31     <section id="portfolio">
32         <h2>Portfolio</h2>
33         <div class="gallery">
34             
35             
36             <!-- Add more images here -->
37         </div>
38     </section>
39
40     <footer>
41         <div class="contact">
42             <h3>Contact Us</h3>
43             <p>Email: example@example.com</p>
44             <p>Phone: 123-456-7890</p>
45             <!-- Add more contact information here -->
46         </div>
47         <div class="copyright">
48             <p>© 2024 MR. Barber. All rights reserved.</p>
49         </div>
50     </footer>
51
52     <script src="js/script.js"></script>
53 </body>
54 </html>

```

Figure 10. Generated index.html (generated by ChatGPT)

6.4 Styling and Functionalities

I decided to simplify the input to help the language model to understand better. Therefore, I went for a minimalist design strategy for the website styling, focusing on a user-centered experience and giving priority to a mobile-first design philosophy for consistent responsiveness on different devices. In Figure 11, you can see content of the style.css, which was generated within seconds, and it is satisfactory.

Additionally, I wanted to integrate smooth scrolling into the website. Luckily, ChatGPT offered a helpful code snippet for this task as you can see the generated JavaScript code in Figure 12, which was both effective and quick. This feature is highly beneficial for beginner developers looking to improve their skills and speed up development workflows.

```

1  /* Resetting default browser styles */
2  body, h1, h2, h3, p, ul, li {
3      margin: 0;
4      padding: 0;
5  }
6
7  body {
8      font-family: Arial, sans-serif;
9      line-height: 1.6;
10     background-color: #f4f4f4;
11     color: #333;
12 }
13
14 .container {
15     max-width: 960px;
16     margin: 0 auto;
17     padding: 20px;
18     box-sizing: border-box;
19 }
20
21 header {
22     background-color: #333;
23     color: #fff;
24     padding: 20px;
25     text-align: center;
26 }
27
28 nav ul {
29     list-style: none;
30 }
31
32 nav ul li {
33     display: inline;
34     margin-right: 20px;
35 }
36
37 nav ul li a {
38     color: #fff;
39     text-decoration: none;
40 }
41
42 section {
43     padding: 20px;
44     margin-bottom: 20px;
45     background-color: #fff;
46     border-radius: 5px;
47 }
48
49 h2 {
50     margin-bottom: 10px;
51 }
52
53 .gallery {
54     display: grid;
55     grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
56     grid-gap: 20px;
57 }
58
59 .gallery img {
60     width: 100%;
61     height: auto;
62     border-radius: 5px;
63 }
64
65 footer {
66     background-color: #333;
67     color: #fff;
68     padding: 20px;
69     text-align: center;
70 }
71
72 footer .contact {
73     margin-bottom: 20px;
74 }
75
76 footer .contact h3 {
77     margin-bottom: 10px;
78 }
79
80 @media (max-width: 768px) {
81     .container {
82         padding: 10px;
83     }
84 }

```

Figure 11. Generated style.css (generated by ChatGPT)

```

js > JS script.js > ...
1  // Smooth scrolling
2  document.querySelectorAll('a[href^="#"]').forEach(anchor => {
3      anchor.addEventListener('click', function (e) {
4          e.preventDefault();
5
6          const target = document.querySelector(this.getAttribute('href'));
7          const offsetTop = target.offsetTop;
8
9          window.scrollTo({
10             top: offsetTop,
11             behavior: 'smooth'
12         });
13     });
14 });
15

```

Figure 12. JavaScript code (generated by ChatGPT)

6.5 Final look and Conclusion

The final appearance of the website is satisfactory as you can see it Figure 13. It adheres to the mobile-first approach and boasts a modern, stylish design. While a UI designer could potentially enhance its aesthetics further, given that it was crafted based on chat input, the outcome is acceptable. The only manual intervention I made was to address the absence of images, as GPT-3.5 cannot generate visual content such as haircut images for the Portfolio section.

Looking ahead, I suggest utilizing Copilot or another AI language model integrated within IDEs for future projects. While ChatGPT adequately facilitated the creation of a simple website, it may not be suitable for more complex developments. Despite its limitations, I thoroughly enjoyed collaborating with ChatGPT to generate content and gain insights into its capabilities. It's evident that for individuals seeking quick guidance on website creation, ChatGPT can offer valuable assistance and examples.

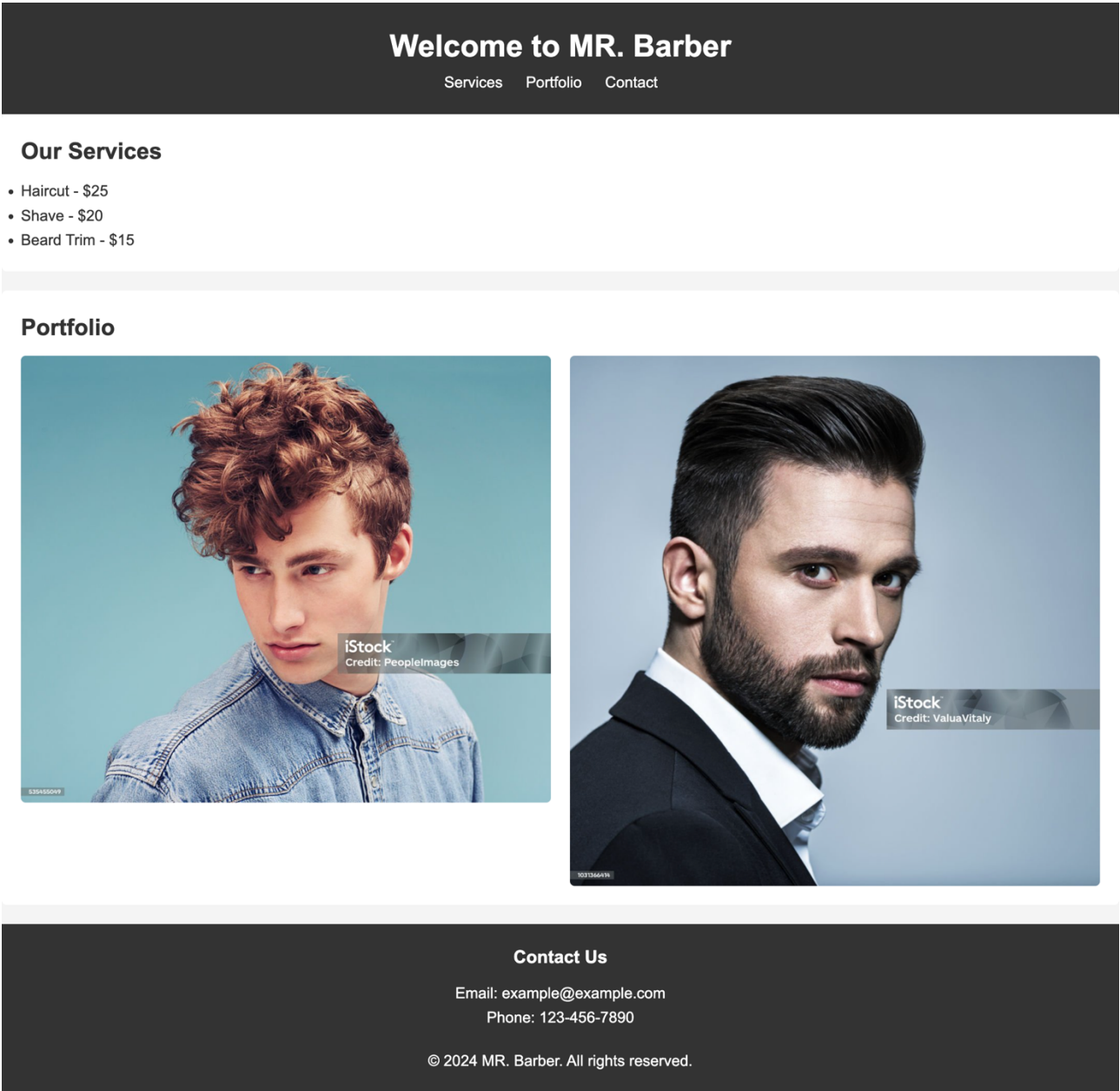


Figure 13. Final look of the generated website

7 Conclusion

The infusion of Artificial Intelligence (AI) into software engineering practices signifies a groundbreaking shift in software development and maintenance methodologies. This dissertation has meticulously examined the diverse roles that AI, notably tools like ChatGPT, play in reshaping software development processes. By exploring the historical context, theoretical foundations, and pragmatic applications of AI, we have unveiled the significant influence these technologies wield in augmenting the productivity, efficiency, and overall capabilities of software engineers.

The research journey encompassed a thorough investigation into the evolutionary trajectory of AI, from its origins to the sophisticated deep learning and natural language processing techniques prevalent today. Landmark achievements in AI history, such as the groundbreaking contributions of Alan Turing and John McCarthy, have paved the way for contemporary advancements in AI, particularly within the realm of software development.

The practical segment of the study focused on integrating ChatGPT into various phases of the software development life cycle. Through quality assurance testing and the creation of a basic website for a barber shop, we illustrated the tangible benefits and implications of employing AI tools in real-world contexts. The outcomes underscored ChatGPT's efficacy in code generation, documentation provision, and debugging assistance, thereby expediting development processes and augmenting efficiency.

An analysis of survey data further validated these observations, revealing that a notable cohort of developers view ChatGPT as a valuable asset that enhances their productivity and job satisfaction. The tool's capacity to reduce time expenditures on coding tasks and bug rectification underscores its potential to streamline routine software engineering endeavours.

Furthermore, we outlined optimal strategies for harnessing ChatGPT in software development, accentuating the significance of comprehending its limitations, selecting pertinent training data, and upholding ethical and responsible usage. These guidelines are pivotal in optimizing the utility of AI tools while upholding precision and relevance in the generated outcomes.

In conclusion, the integration of AI, particularly language models like ChatGPT, into software engineering methodologies holds vast promise for revolutionizing the discipline. Despite existing challenges and constraints, the advantages in terms of productivity, efficiency, and enriched proficiencies for software engineers are considerable. As AI technologies continue to progress, further exploration and practical deployments will be imperative to fully exploit their transformative capabilities. This dissertation furnishes a comprehensive insight into AI's pivotal role in software

development, laying the groundwork for future innovations and implementations in this dynamic and swiftly evolving domain.

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